

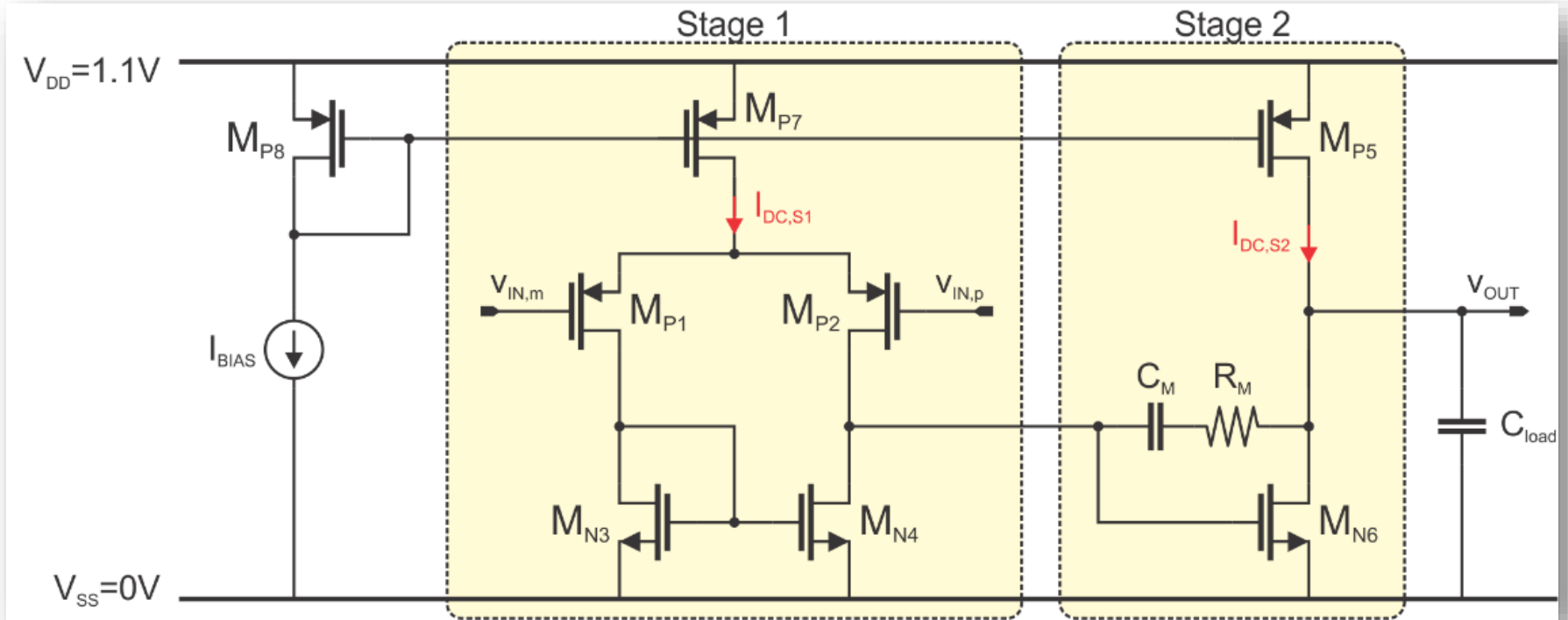


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Analog Electronics – Final Project

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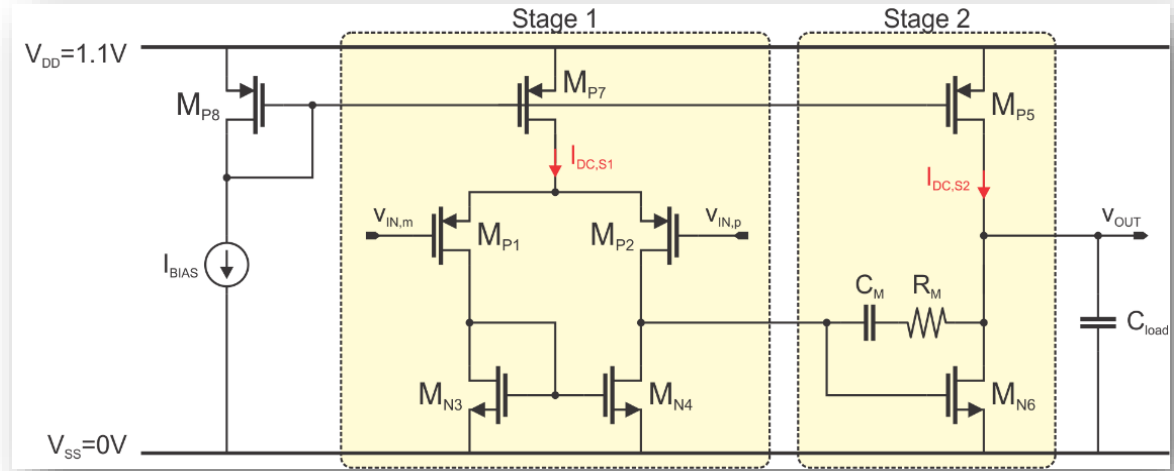
2-Stage (Single-Ended) Miller-Compensated OTA



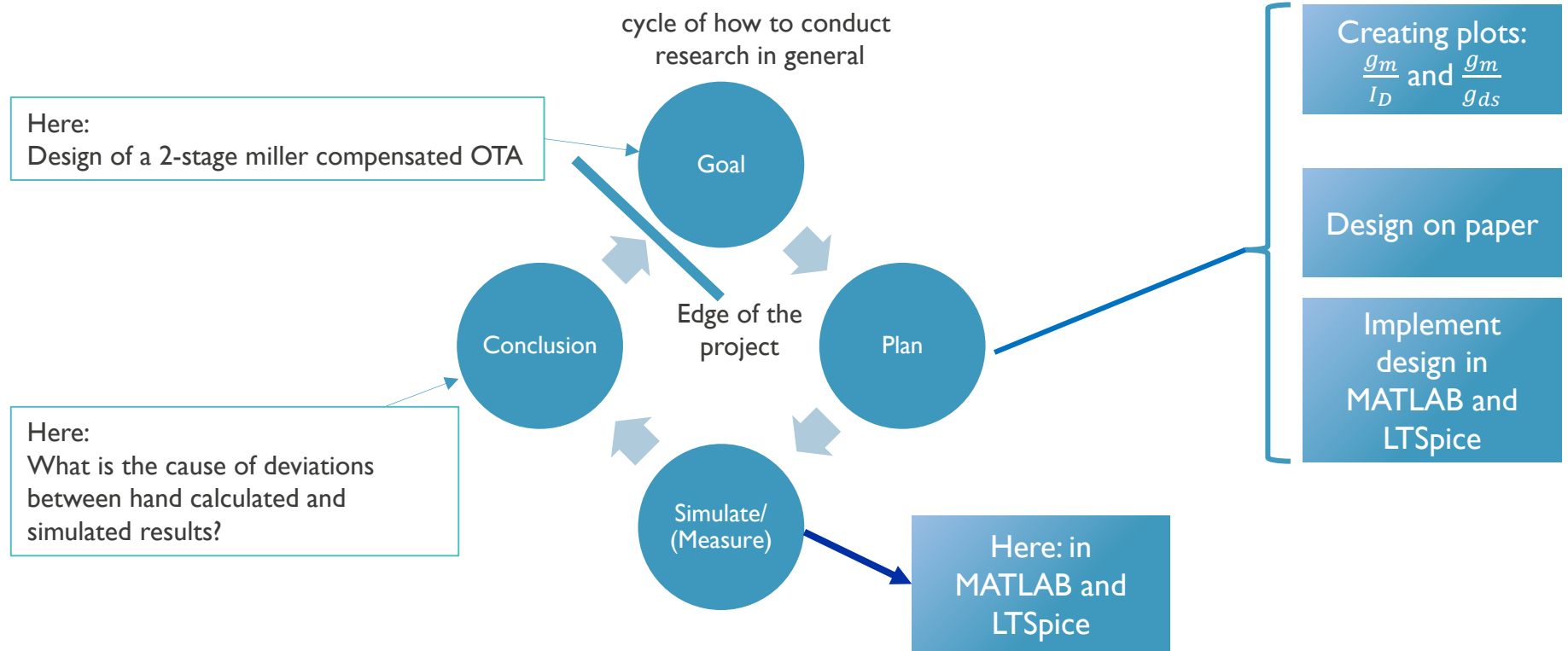
Final project

Overview

- Almost 3 full sessions:
 - 04.12.2025 (3h)
 - 11.12.2025 (4h)
 - 18.12.2025 (4h)
- 2 Goals:
 - 1) Design of a 2-stage miller compensated OTA
 - 2) Fill out the “Design Project Report”
 - deadline: 2 weeks before oral exam? (to be confirmed)
- Quality of your report will impact your final grade of this class
 - not delivering the report before deadline or giving an empty report will lead to **failure** of this class!



You are the researcher now!



Final project

Overview of today's session

- each section gives 10 points
- How to fill the report?
 - Follow the instructions listed in this report
 - Hints and remarks help you to speed-up your design progress

➔ **Focus on having working MATLAB-code today!**
Ask the TAs for help

1st session goal

Your goal from
next week till deadline

- Group data
- Goal:
- Plan: Design of the 2-stage OpAmp on paper (10 points)
- Design: Implementation of the OpAmp in MATLAB and LTSpice (10 points):
- Experiment (10 points):
- Conclusion (10 points):
- Appendix:

New This Year

- Minimum width of 1 μm to try to minimize narrow-channel effects that are not modelled in the MATLAB library
 - This could lead to problems for some specifications, so let us know if you actually hit this limit
 - If you hit this limit, you should most of the time be able to just increase L
- LTspice simulations at multiple temperatures + discussion

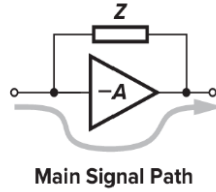
Final project

Overview of the coming slides

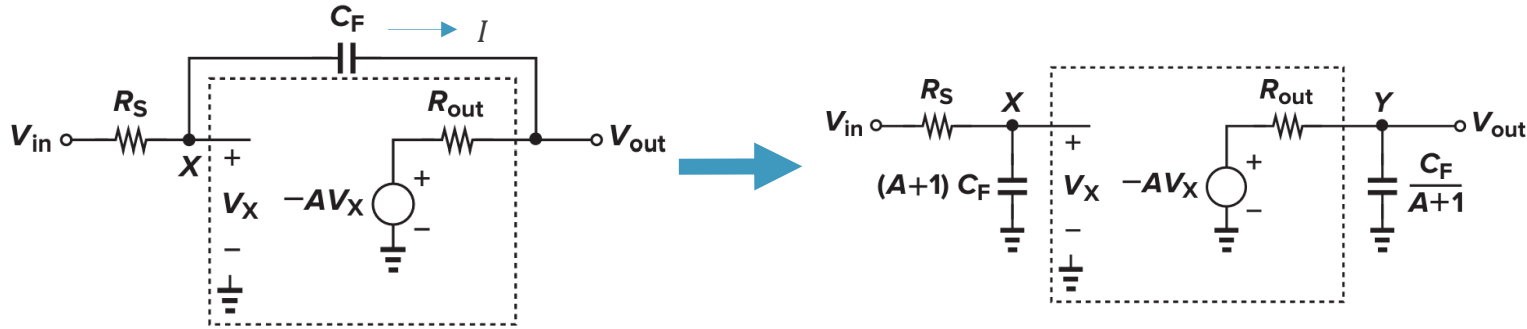
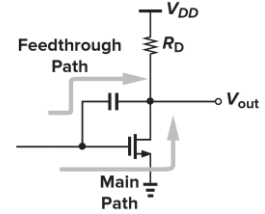
1. Reminder: some key concepts in Miller OTA and how to approach them
 - Miller's Theorem
 - Poles & zeros: derivation, concept of pole splitting, pole-zero doublet
 - Nulling Resistor: purpose & derivation
 - Current mirroring
2. Quick overview: how to design to the specifications
3. Content of the "Design Project Report"
4. LTspice guidelines

I. Reminder

Miller's Theorem



■ Typical case for valid application of Miller's theorem



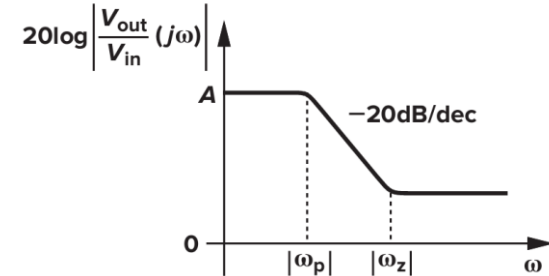
$$I = \frac{V_{in} - V_X}{R_S} \rightarrow 1. V_{out} = -AV_X + IR_{out}$$

$$2. I = (V_X - V_{out})C_F S$$

$$\frac{V_{out}}{V_{in}}(s) = \frac{R_{out}C_F S - A}{[(A+1)R_S + R_{out}]C_F S + 1}$$

Zero and poles are at:

$$\begin{aligned} - \omega_Z &= \frac{A}{R_{out}C_F} = \frac{g_m}{C_F} \\ - \omega_p &= - \frac{1}{[(A+1)R_S + R_{out}]C_F} \end{aligned}$$

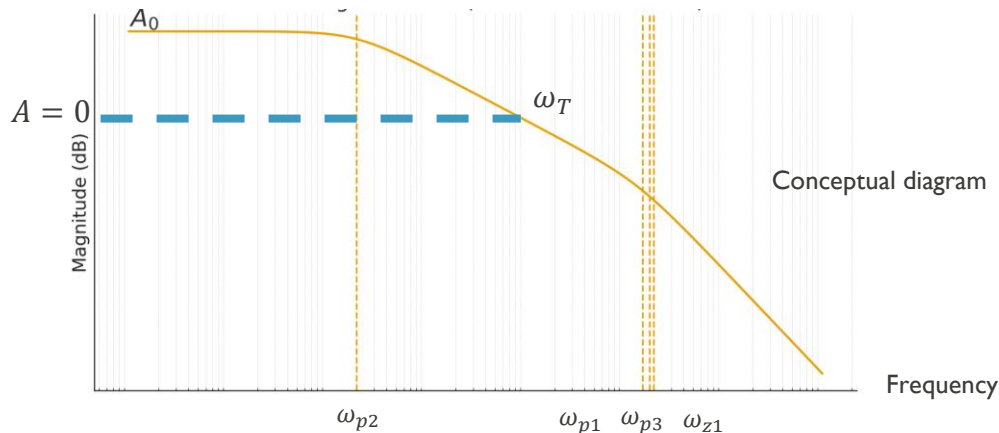


- $$\text{pole frequency at node } n \text{ in rad/s} = \frac{1}{\text{Total resistance between node } n \text{ and AC ground}} * \frac{1}{\text{Total capacitance between node } n \text{ and AC ground}}$$

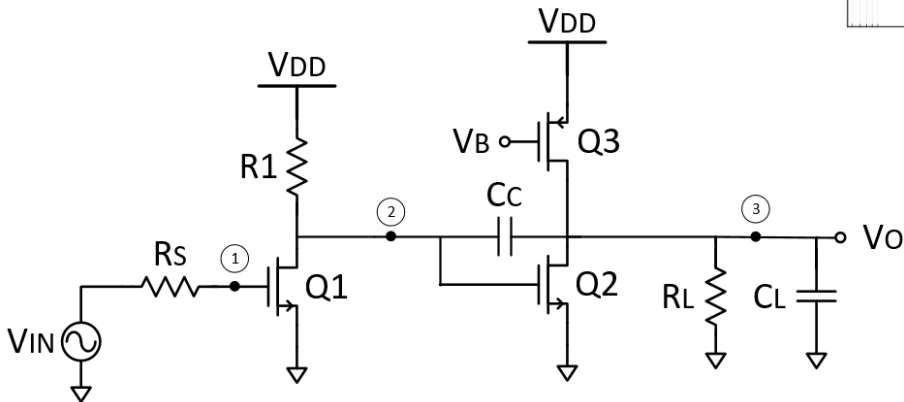
Example circuit with zero + poles

Finding zero + pole frequencies

- $$\omega_{p1} = \frac{1}{R_S * (C_{gs1} + C_{gd1} * (1 + g_{m1} * R_1))}$$
- $$\omega_{p2} = \frac{1}{R_1 * (C_c * g_{m2} * R_L)}$$
- $$\omega_{p3} = \frac{g_{m2}}{C_L}$$
- $$\omega_{z1} = \frac{g_{m2}}{C_c}$$



$$A_0 = 20 * \log \left(\frac{\omega_T}{\omega_{p2}} \right)$$



Pole splitting

- Increasing C_{gd} pushes the dominant pole to lower frequencies and the non-dominant pole to higher frequencies
 - For small C_{gd} , gate and drain nodes are separated
 - For big C_{gd} , $A_0 C_{gd}$ dominates the gate node capacitance, and defines the dominant pole
 - Gate and drain are shorted by big C_{gd} at the frequency of the non-dominant pole $\rightarrow$ output resistance of Mn becomes $1/g_m$

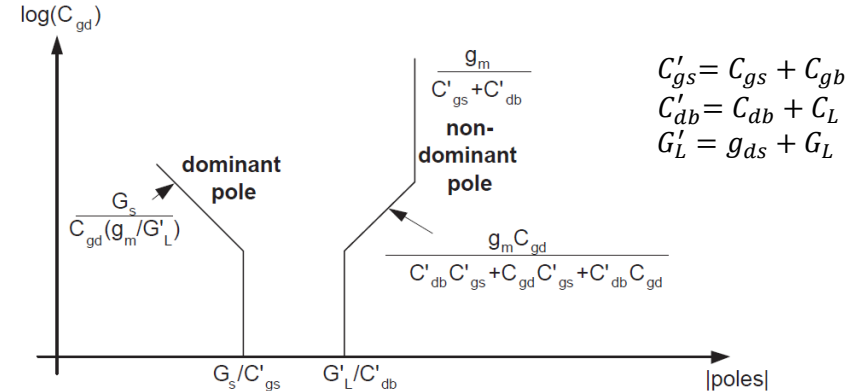
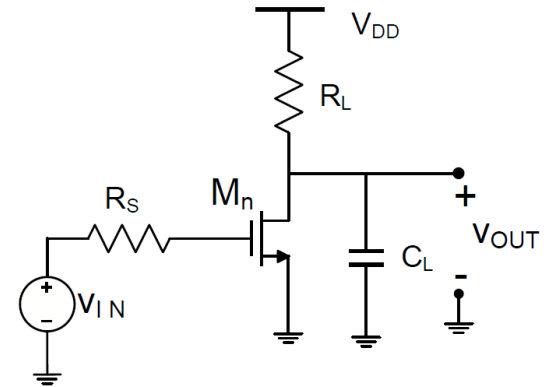
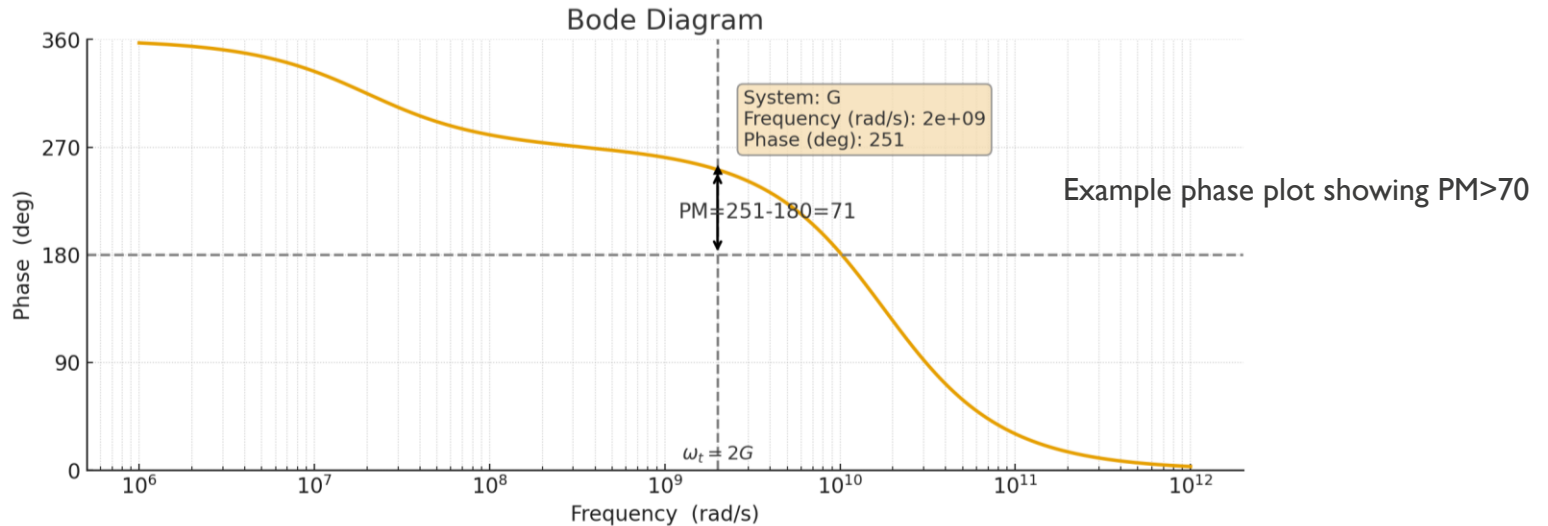


Figure 2.15: Bode diagram of pole splitting: asymptotes indicate the position of poles as a function of C_{gd} on a log-log scale.

Phase Margin

- $Total\ phase\ shift = 0 - \tan^{-1} \frac{\omega_T}{\omega_{p2}} - \tan^{-1} \frac{\omega_T}{\omega_{p1}} - \tan^{-1} \frac{\omega_T}{\omega_{p3}} - \tan^{-1} \frac{\omega_T}{\omega_{z1}} = -\phi$
- $Phase\ margin = -\phi - (-180^\circ) > 70^\circ$



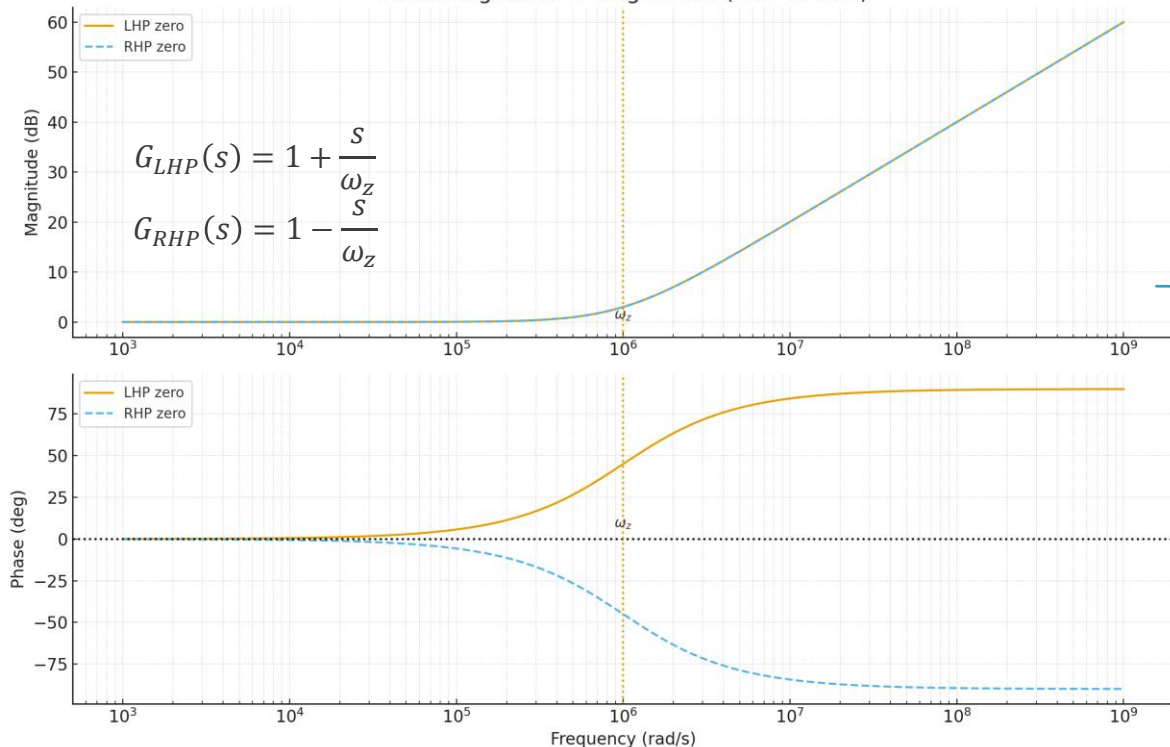
Problem with a RHP zero

- Why is it a problem?

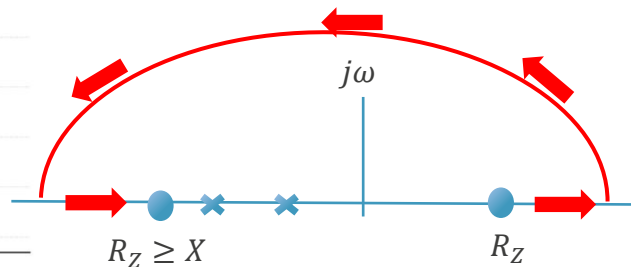
Because it boosts the magnitude but lags the phase!

Worst possible combination for stability

Bode Diagram of a Single Zero (LHP vs RHP)



Solution:



X is an expression you should find for your OTA.

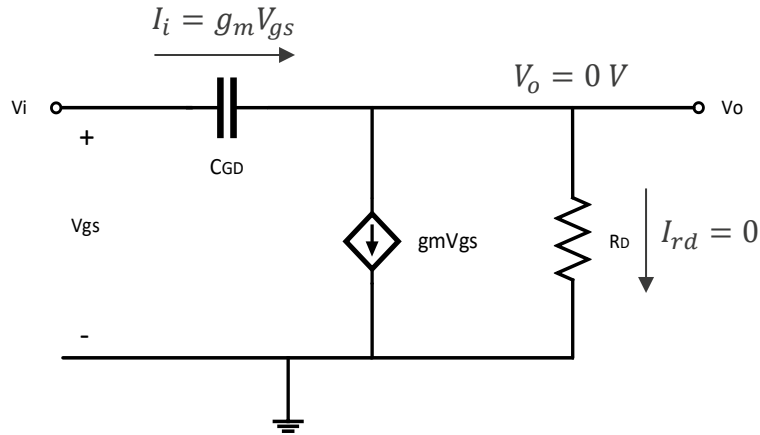
You either:

- Eliminate zero
- LHP zero

The pole should be chosen so that GBW stays where you want it (*not* to place the pole to satisfy PM constraints)

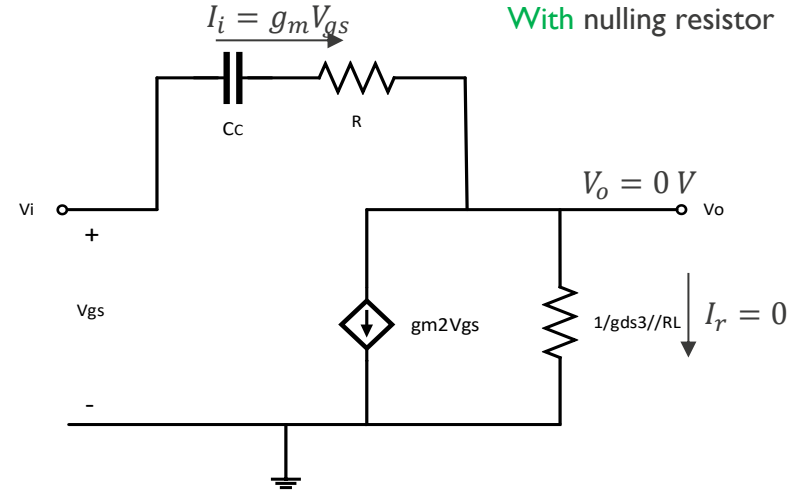
Expression of the zero

Without nulling resistor



- $\frac{V_i}{1/(s_Z * C_{GD})} = g_m * V_i$
- $s_Z = \frac{g_m}{C_{GD}}$

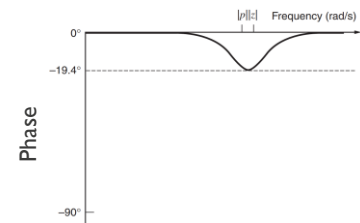
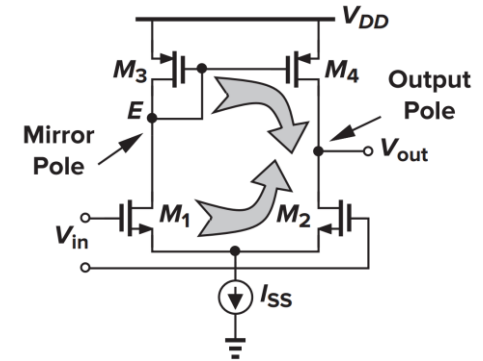
With nulling resistor



- $\frac{V_i}{R + \frac{1}{s_Z * C_c}} = g_{m2} * V_i$
- $s_Z = \frac{1}{C_c * (\frac{1}{g_{m2}} - R)}$

Pole-zero doublet

- Doublet is a pair of a zero and a pole that are at slightly different frequencies
 - Occurs when pole cancellation is not perfect
 - Occurs in a differential amplifier with an active load
 - It affects the step response of the amplifier
- Non-dominant pole associated with node E: $\omega_{p,E} = \frac{g_{m3}}{C_E}$
 - C_E is the total capacitance of the node E
- A LHP zero occurs because there are two paths to the output
 - $\omega_z \approx 2\omega_{p,E}$



FINAL REMARK

$|p_2|/GBW_T$ vs Phase Margin in Two-poles System

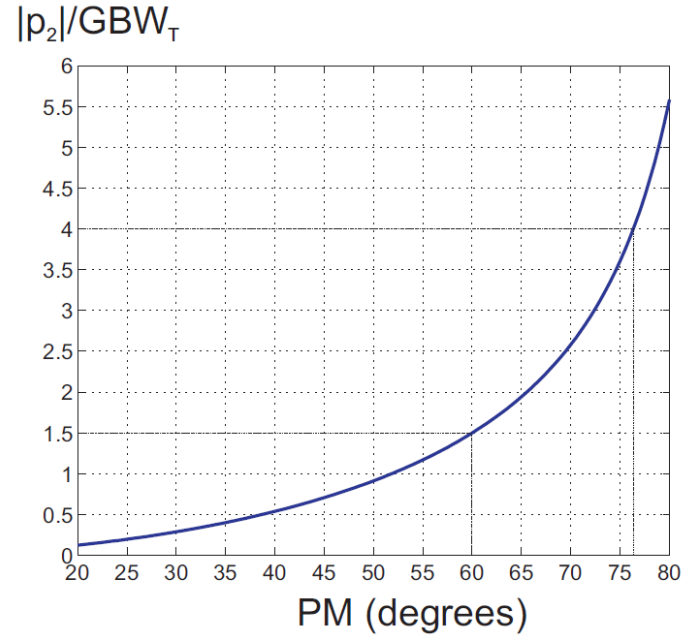
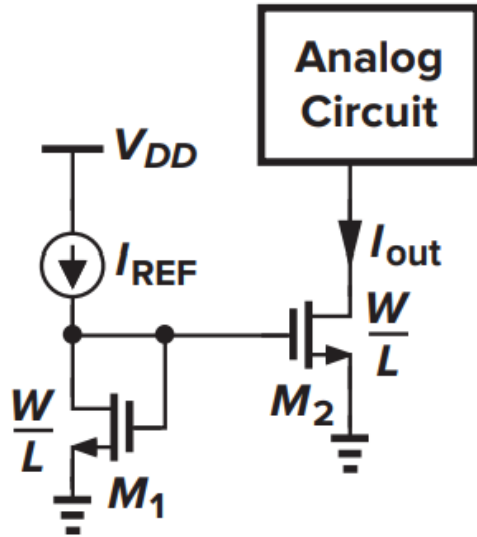


Figure 7.3: Relation between the phase margin and the ratio of the second pole and GBW_T .

Current Mirror



Current Mirror Architecture

$$I_{REF} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{GS} - V_{TH})^2$$

$$I_{out} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_2 (V_{GS} - V_{TH})^2$$

$$I_{out} = \frac{(W/L)_2}{(W/L)_1} I_{REF}$$

Ignore channel length modulation.

2. How to design to specifications

How to start

- Think High-Level: Gain distribution between stages?
- Write down the defining equations of your specifications
- Derive the expression of the dominant pole and the gain of your different stages
- Start from the output stage
 - PM constraint limits g_{m6} in function of GBW (analogous to session 2, especially final slide of session 2)
 - Choose lengths and V_{ovs} for M_{n6} and M_{p5} based on purpose (gain [depending on gain distribution] and current source function respectively). Use plots for this.
- Fill in the definition of GBW
 - Most things cancel out, leaving an expression for g_{m2}

General Remarks

- V_{ds} is often a free choice
 - Ideal choice is based on min and max values to stay in saturation for that stage
 - Watch out: sometimes v_{ds} is based on v_{gs} somewhere else
- Look carefully at $M_{n3,4}$ and M_{n6} . You can't choose both V_{ov} values independently
 - $V_{gs3}=V_{ds3}=V_{gs4}=V_{ds4}=V_{gs6}$
- I_{ds8} is a completely free choice, but choose a reasonable value (for example identical to the smallest current mirror transistor)
- For MATLAB, only write the numbers that you CHOOSE and not the ones you calculate or read from a figure (MATLAB can do that better for you). The rest should be inserted based on the formulas you used.
 - Choices consist mostly of V_{ovx} , V_{dsx} and L_x

3. Content of the “Design Project Report”

Design Project Report

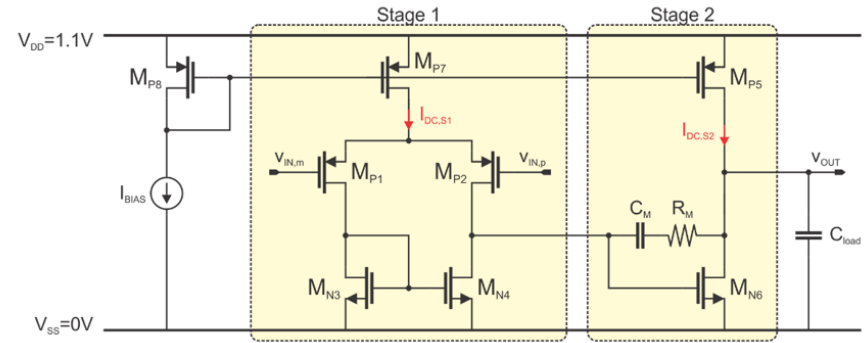
In the following each section is briefly discussed

- Group data
- Goal:
- Plan: Design of the 2-stage OpAmp on paper (10 points)
- Design: Implementation of the OpAmp in MATLAB and LTSpice (10 points):
- Experiment (10 points):
- Conclusion (10 points):
- Appendix:

Design Project Report

Setup your goals by inserting your specification

Goal:

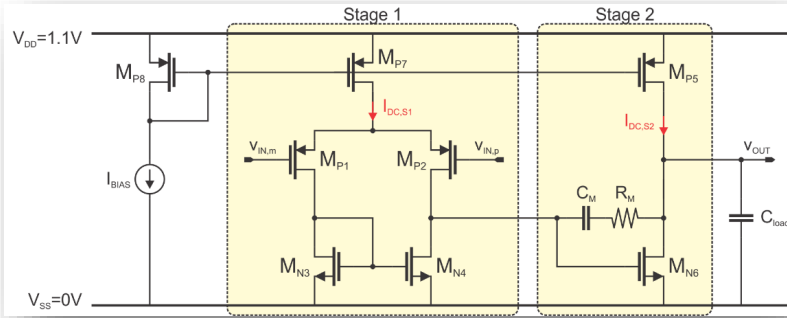


Design of the 2stage OpAmp such that it passes the given specifications (insert your specs below):

C_{load} [pF]	
DC gain [dB]	
f_{GBW} [MHz]	
Phase margin (PM) [deg]	> 70
Output swing [V]	> 0.7

Design Project Report

Design on paper (I)



Create your plots, as presented during 2nd session – CS with active load

Write down all your equations you used to calculate the small signal parameters step-by-step. Scans of handwritten derivation is also allowed.

All Transistors must be in saturation !

Plan: Design of the 2-stage OpAmp on paper (10 points)

- Calculate the V_{OV} , g_m , I_{DS} and L of each transistor (see exceptions under "Remark") and the required C_M and R_M in the OpAmp circuit. For that, insert all your calculations as well as your $\frac{g_m}{g_{ds}}$, $\frac{g_m}{I_D}$ -plots you used for your handcalculations below:

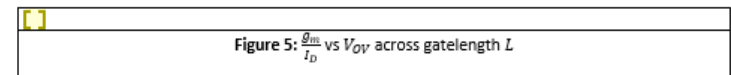
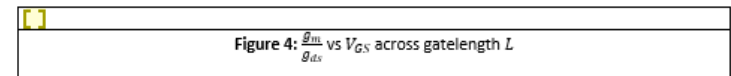
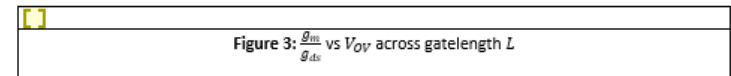
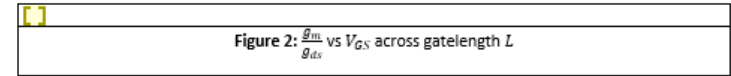
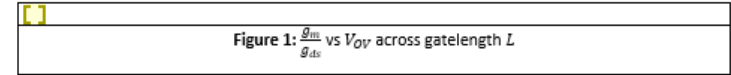
Hints:

- First, plot $\frac{g_m}{g_{ds}}$ and $\frac{g_m}{I_D}$ across V_{OV} and gatelength L and do it again across V_{GS} , as presented in the 1st session. Think about whether to create the plots for a PMOS or for a NMOS device.
- Then, based on those plots, start your calculations.
- Furthermore, you can assume the following:
(1) the OpAmp is designed in triple-well-technology (i.e. $V_{SB} = 0$); (2) $V_{dsat} \approx V_{OV}$ and $V_{dsat} \geq 50mV$ across V_{OV} ; (3) $C_M = \frac{1}{4} \cdot C_L$

Remarks:

- For M_{P5} , M_{P7} and M_{P8} you are not required to calculate the g_m

Plots used for the handcalculations:



Design Project Report

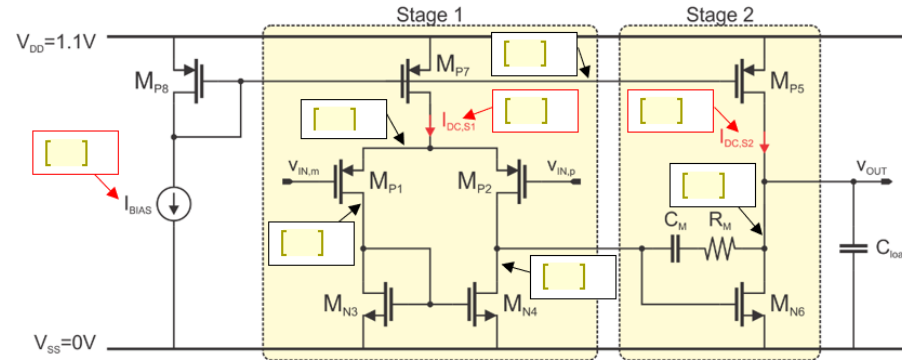
Design on paper (2)



not just numbers, but
give also the entire equation

- 2) Place all the calculated voltages (in the blackbox) and all currents (in the red box) on the circuit depicted below and calculate also:

$V_{cm,in,min}$	[]
$V_{cm,in,max}$	[]
$V_{out,min}$	[]
$V_{out,max}$	[]
P_{diss}	[]



Design Project Report

Design

All (relevant) values should be identical to the handcalculations.
Otherwise, the difference must be explained in the handcalculations.

final_project.m

```

1 final_project.m
2
3 % Project: circuit
4
5 % Parameters
6 % =====
7 % Input: analog, circuitFile, top, elementList, spec, choice, ...
8 % Output: designClass, RVOT, RVOT, simulator, simuFile, simuFileFile
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10 % Check if choice is valid
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```

Design: Implementation of the OpAmp in MATLAB and LTspice (10 points):

- Based on your handcalculations, create your OpAmp design in MATLAB. After completion, please insert your final and entire MATLAB code after the appendix (i.e. at the end of this report-document).

Remark:

- Please make sure that all widths W are below 1mm

- Fill out both tables depicted below. All values are to be determined in MATLAB.

Device sizes and bias point parameters according to MATLAB

Device	W [μm]	L [nm]	I _{th} [μA]	V _{OV} [V]	g _m [S]	g _{ds} [S]	g _m /g _{ds} [-]	V _{DS,sat} [V]	V _{th} [V]
M _{p1}									
M _{p2}									
M _{n3}									
M _{n4}									
M _{p5}									
M _{n6}									
M _{p7}									
M _{n8}									

Device	Units	Value
C _M	pF	
R _M	Ω	
I _{BIAS}	A	

- For those who have finished the code between this class and the next one
 - the other slides of this presentation show how to run the LTspice simulations

Design Project Report (session 2)

Experiment

- Further details about each experiment is explained in the report
 - Ask TAs for help, if instructions are not clear
- After each simulation you are asked to “Explain, analyse and interpret”
 - This shows us that you understand each experiment!
- How to explain, analyse and interpret?
 - Ask questions like
 - what do the results clearly indicate?
 - What have you found?
 - Then, find logical explanations for these results

Experiment (10 points):

In this section you will simulate the following:

- A. Frequency Response in MATLAB and LTSpice
- B. Noise contribution in LTSpice
- C. Linearity in LTSpice

Note on how to present graphs in general:

- When you make a graph, put labels on every axis to make clear what you are showing. Also, show the units!
- When you plot multiple curves on one graph, add a legend.

Design Project Report (session 2)

Conclusion

- This summarizes your research!

Prove that you understand, from where the deviations are coming from

If you are passing all the specs, you can leave this section empty

from section “Plan: Design of the 2-stage OpAmp on paper”

calculate it

Conclusion (10 points):

16) Conclude your experiment by filling the editable fields in the performance table depicted below

Metric	Units	Specification	from hand-calculations	MATLAB	LTspice
DC gain	magnitude	[]	[]	[]	[]
DC gain	dB	[]	[]	[]	[]
Gain-Bandwidth frequency f_{GBW}	MHz	[]	[]	[]	[]
Dominant pole frequency f_{dom}	kHz	[]	[]	[]	[]
PM	°	> 70	[]	[]	[]
$V_{in,cm,max} - V_{in,cm,min}$	V		[]	[]	[]
$V_{out,max} - V_{out,min}$	V	> 0.7	[]	[]	[]
P_{diss}	mW		[]	[]	[]
$\overline{v_{out,eq}^2}$ (output-referred noise)	V _{rms}				[]

17) Comment about the deviations between hand calculation, MATLAB and LTspice values found above.
Conclude, what the causes of such deviations are.

[]

18) If you have results which are not passing the specifications: conclude for each of those result what the cause is and what you need to do in order to make it pass, if you were repeating this experiment.

[]

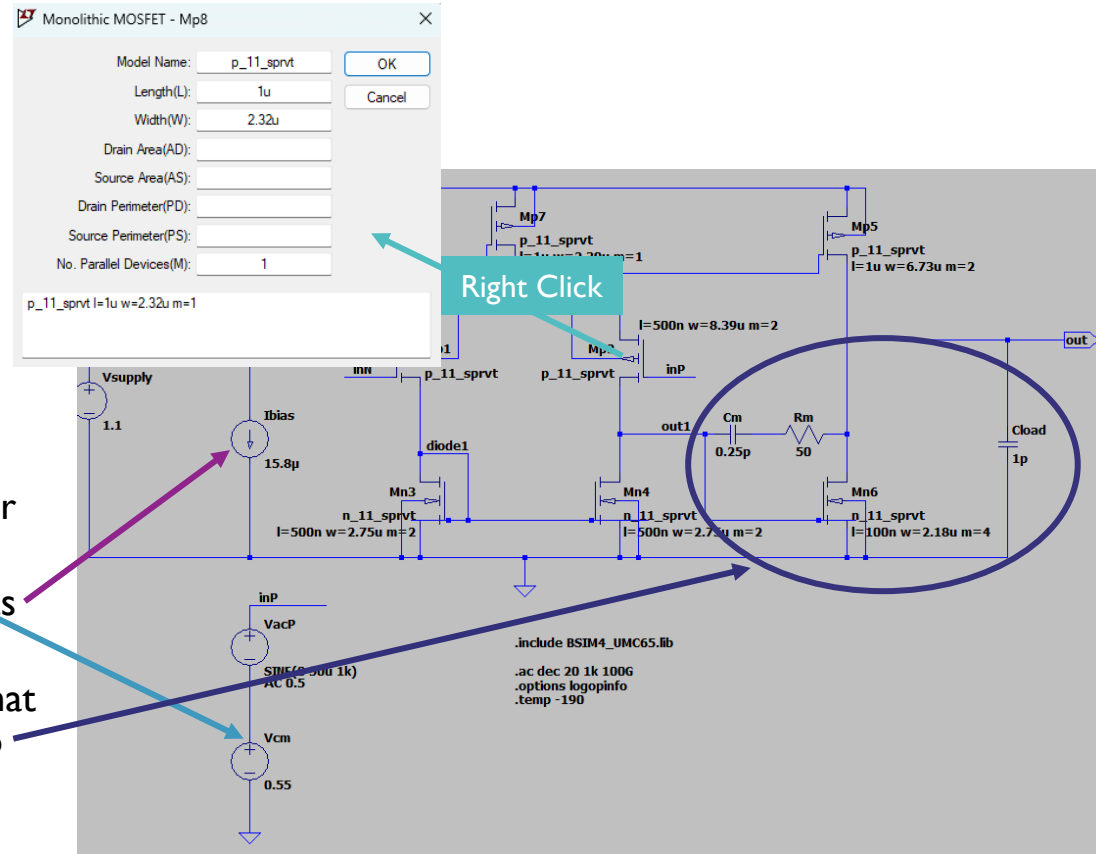
from section “Experiment”

4. LTspice guidelines

LTSPICE

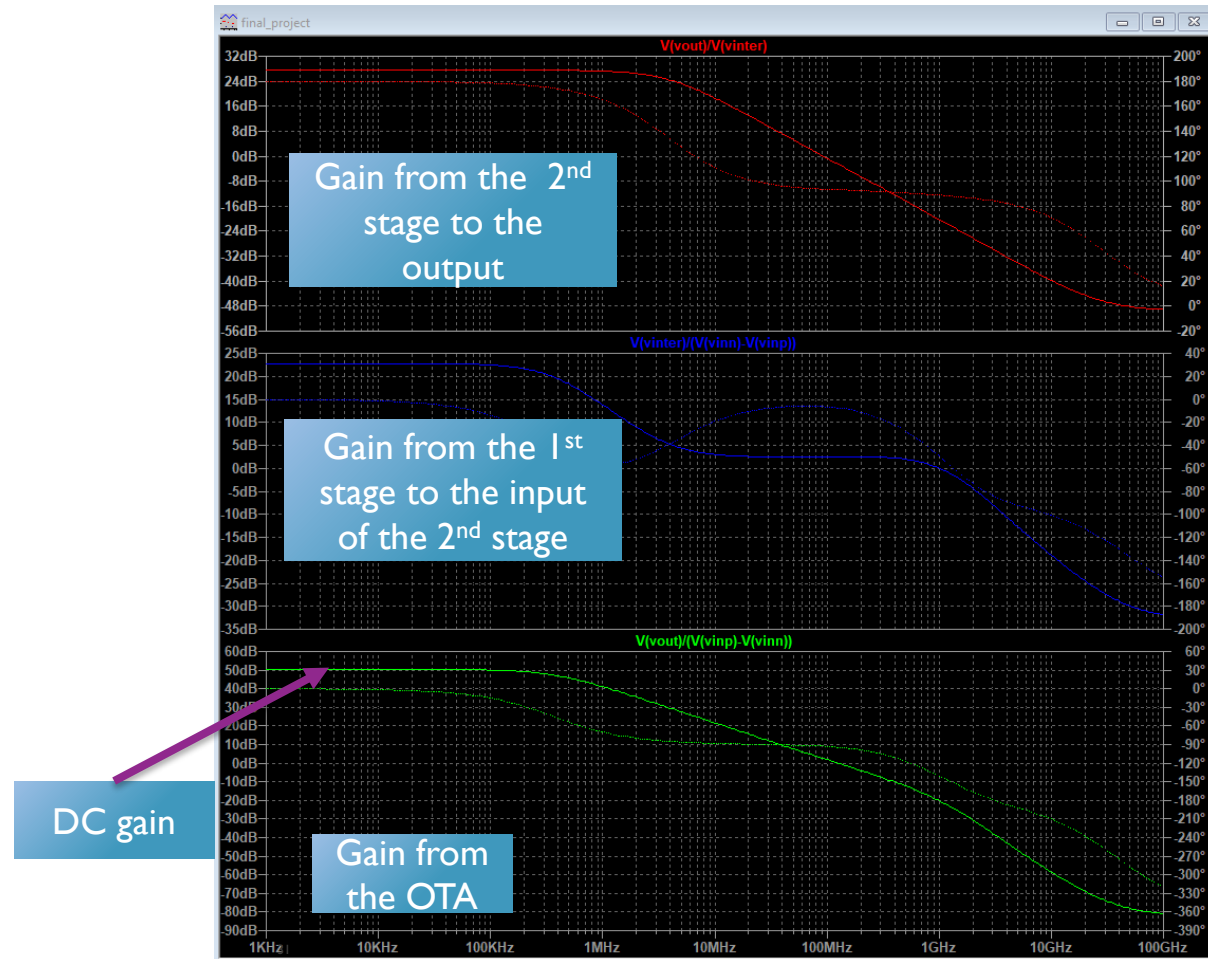
AC plot

- Use the file “final_project”
 - Change the transistors' **W**, **L** and **M**
 - LTspice **W** is your finger width
 - LTspice **L** is your transistor length
 - LTspice **M** is your number of fingers
 - Change the **VCM** DC voltage to your desired output DC voltage
 - Change **IBIAS** to your calculated bias current
 - Change the values of **CLOAD** to what was given to you and **CM** and **RM** to what you calculated
 - Run



LTSPICE

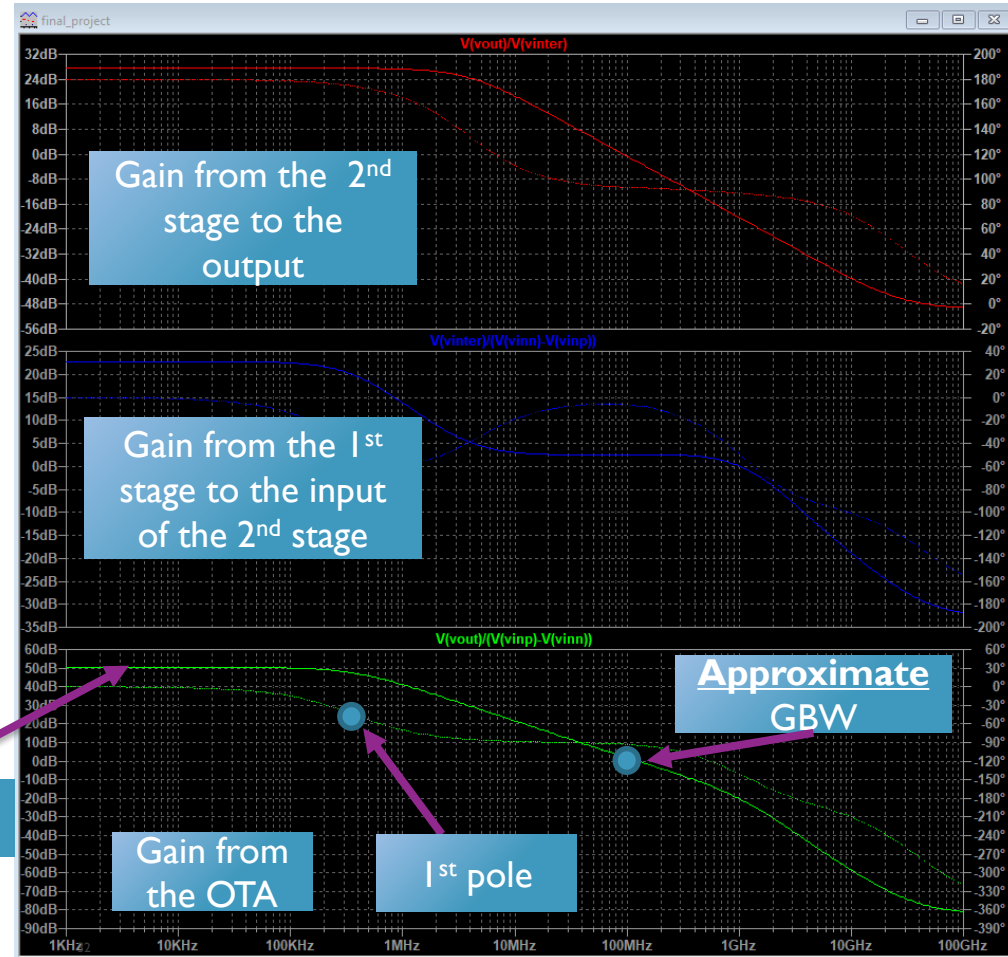
AC plot



LTSPICE

AC plot

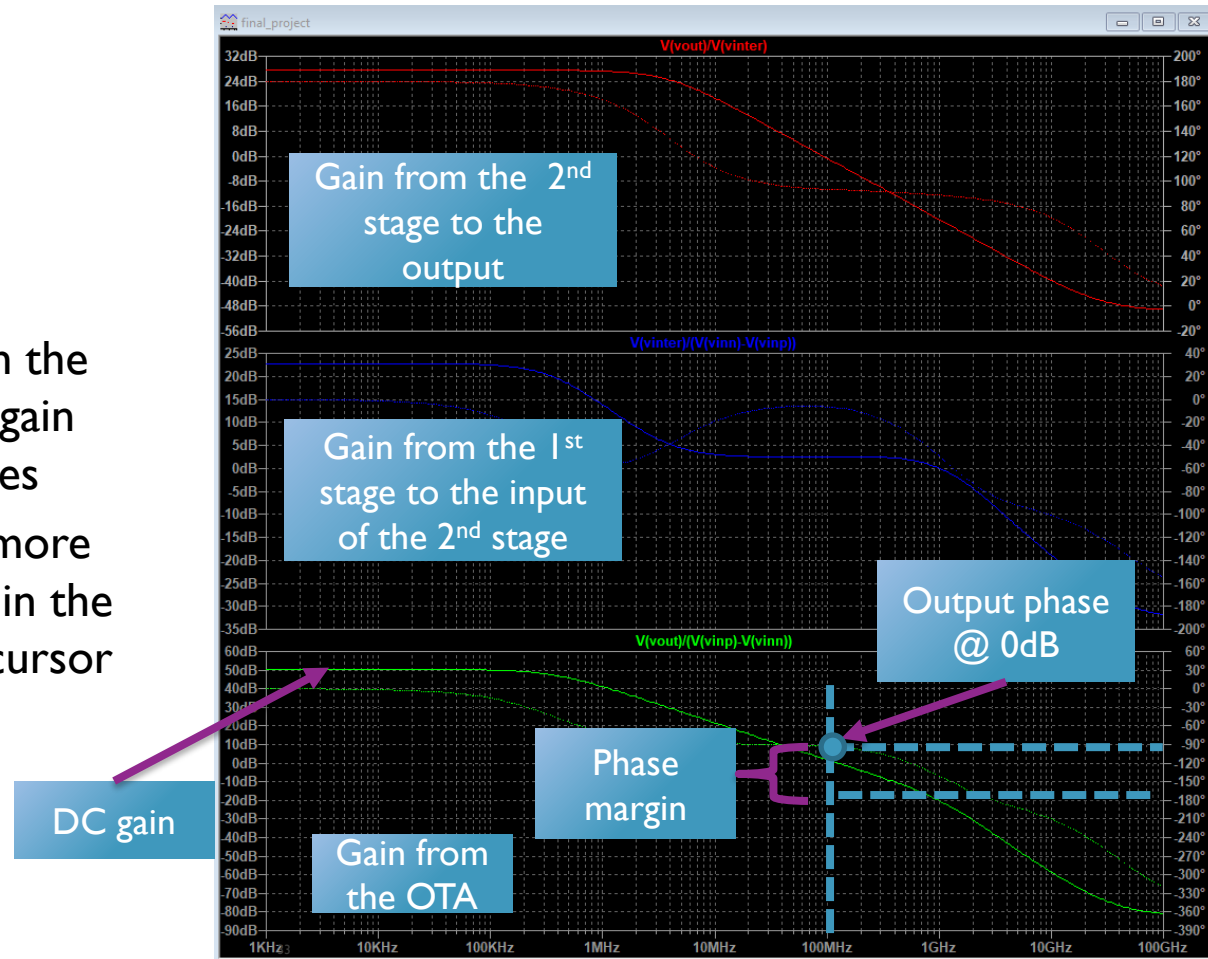
- GBW
 - DC gain (V/V) x 1st pole (Hz)
 - Assuming a 1 pole system
 - 1st pole is where the DC gain fall by 3dB
 - 1st pole is the frequency where phase is 45 (easier to spot visually)
- **Sanity check:** GBW is approximately the frequency @ 0dB (unity gain frequency)
 - Don't use this as the GBW value in the report!



LTSPICE

AC plot

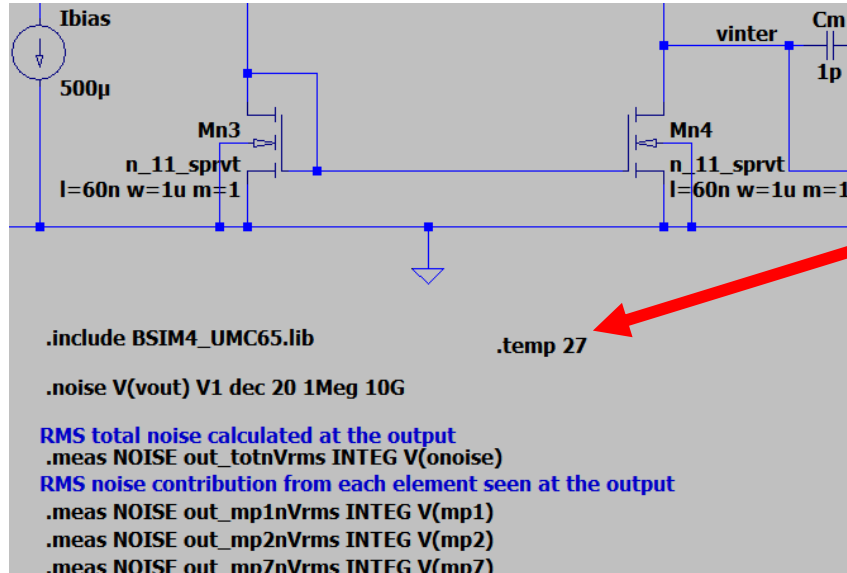
- Phase margin
 - Phase difference between the output phase at the 0dB gain frequency and 180 degrees
 - Use the cursor to get a more precise value (right-click in the graph name and change cursor from “None” to “1st”)



LTspice

Temperature

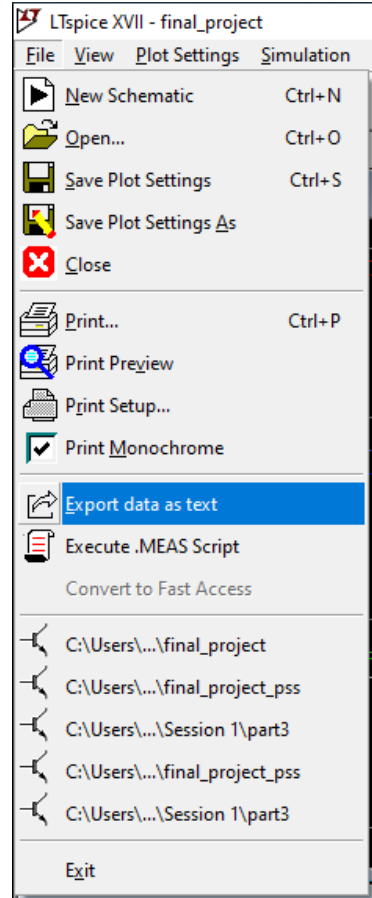
- The .tran spice directive is used to set the simulation temperature
 - Temperatures are given in °C



LTspice

LTspice -> MATLAB

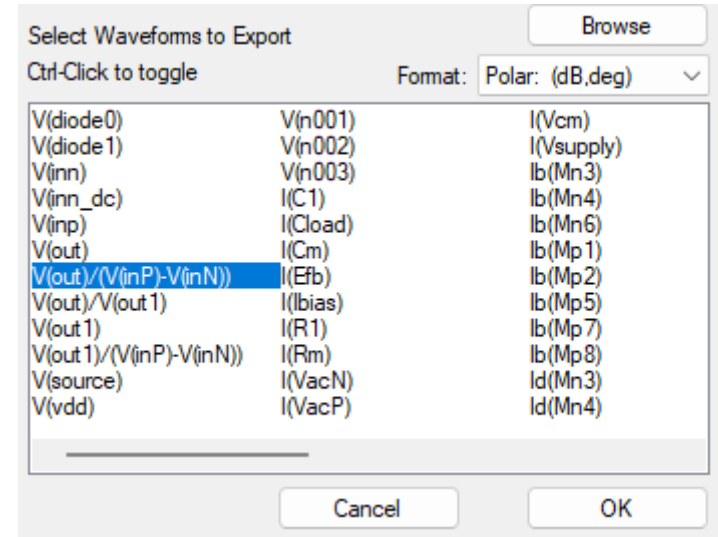
- **Click on the window with the plots** (otherwise the necessary option won't show up)
- Click File -> Export data as text



LTspice

LTspice -> MATLAB

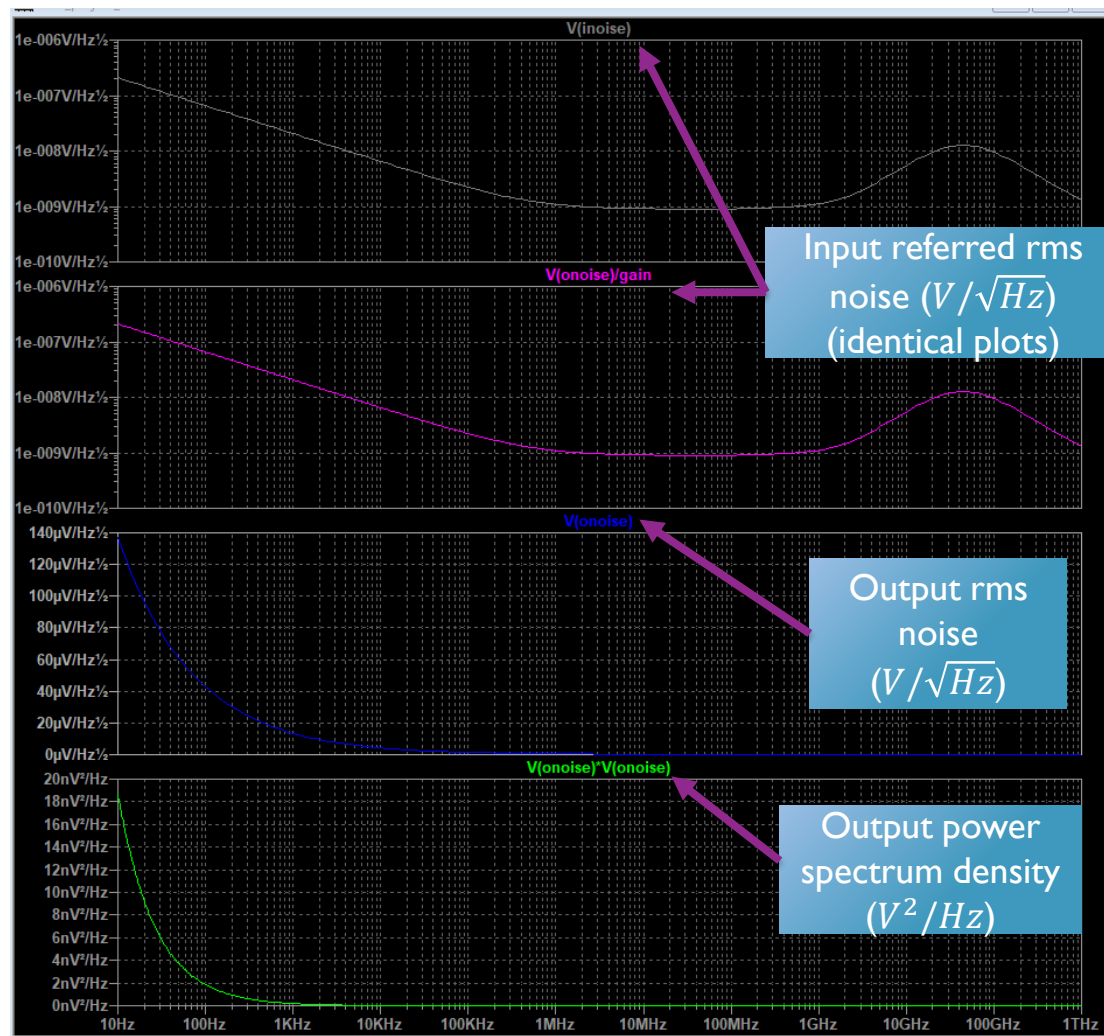
- Select the trace you want and choose the output file name on top
- Keep format as “Polar: (dB,deg)” for the script to work
- To see how to use this file in MATLAB, check “LTspice_example.m” on Canvas in the “Final Project” folder.
 - This uses a MATLAB function “read_LTspice_data” which you can find in the same folder on Canvas.
 - **The MATLAB template for the project already includes plotting code for this.**
 - Use the filename “final_project_data.txt” for this and place it in the same directory as the MATLAB code.



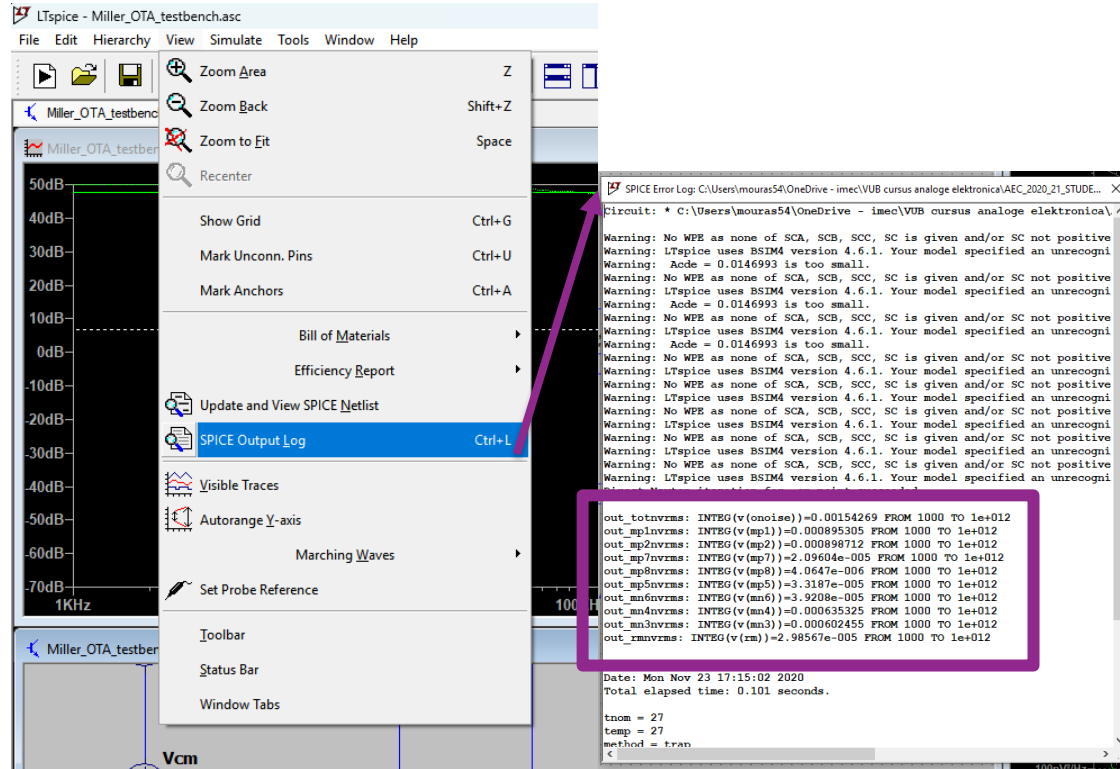
LTSPICE

Noise

- Use the file “final_project_noise”
 - Change the transistors W, L, VCM, CLOAD, CM, RM, IBIAS the same way as done before
 - Run



- Use the file “final_project_noise”
 - To print the integrated noise power contribution of each element at the output, use the command shown in the figure
 - The tab “View” with these options only appears when the circuit window is selected
 - The first line is the total output referred noise in V_{rms} which you need for your report



MAKE SURE YOUR FOCUS IS ON THE SCHEMATIC AND NOT THE PLOT.
OTHERWISE, THE OPTION WON'T SHOW UP

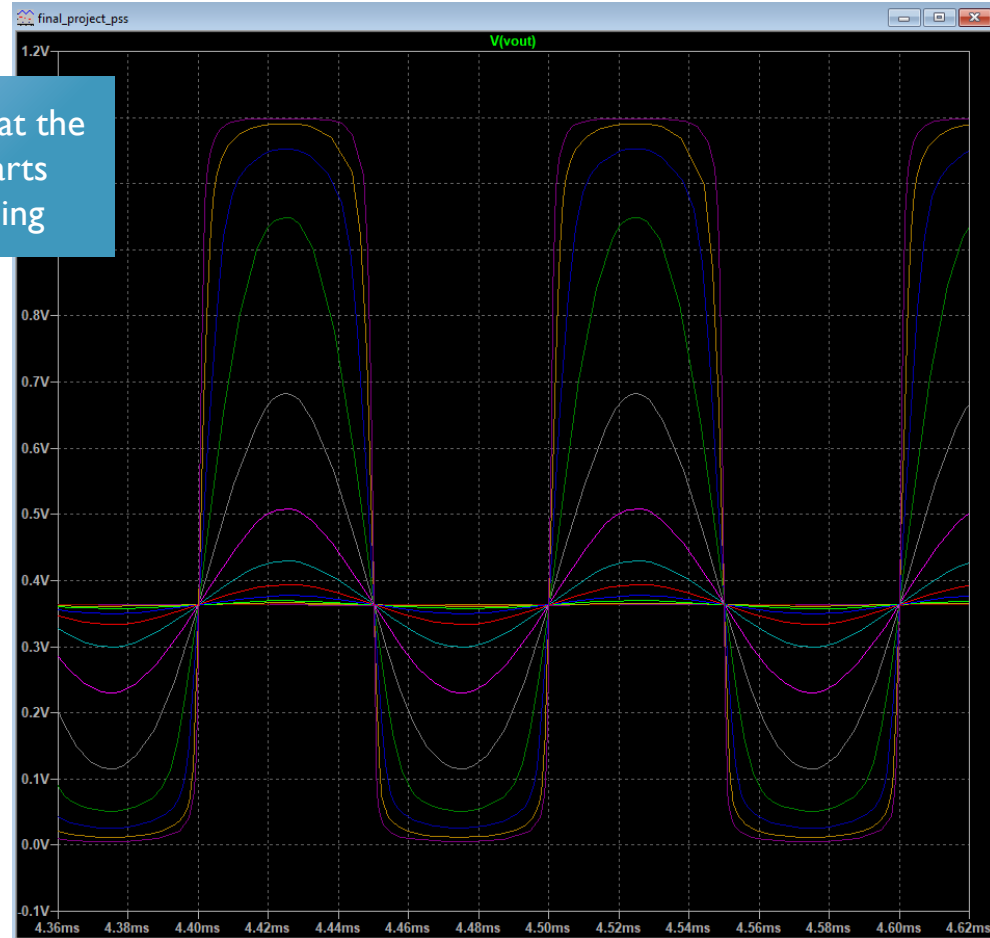
LTSPICE

Compression

NO LONGER REQUIRED

Waveforms at the
output starts
compressing

- Use the file “final_project_pss”
 - Change the W, L, VCM, CLOAD, CM, RM, IBIAS the same way as done before
 - Run (it may take some seconds to finish)

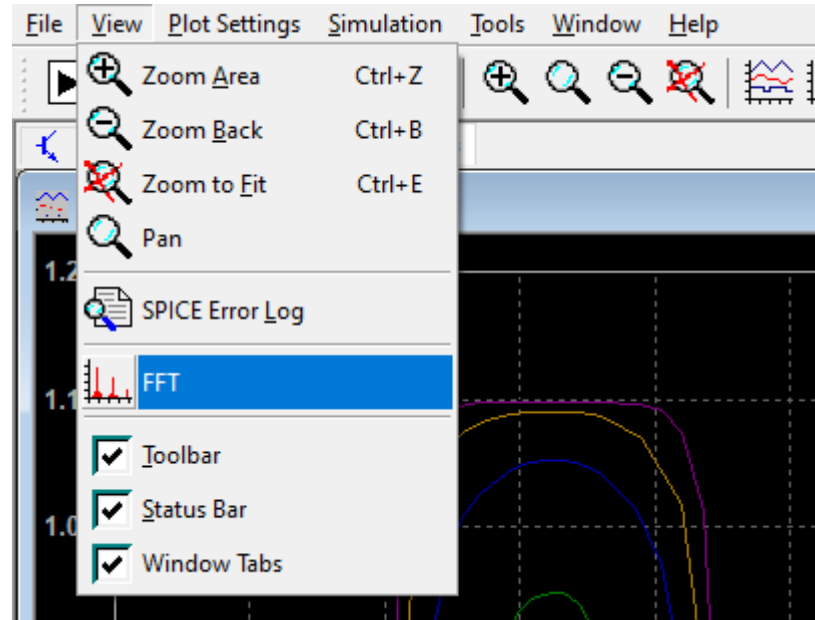


LTSPICE

Compression

NO LONGER REQUIRED

- Use the file “final_project_pss”
 - With the plot window selected, choose the tab View -> FFT

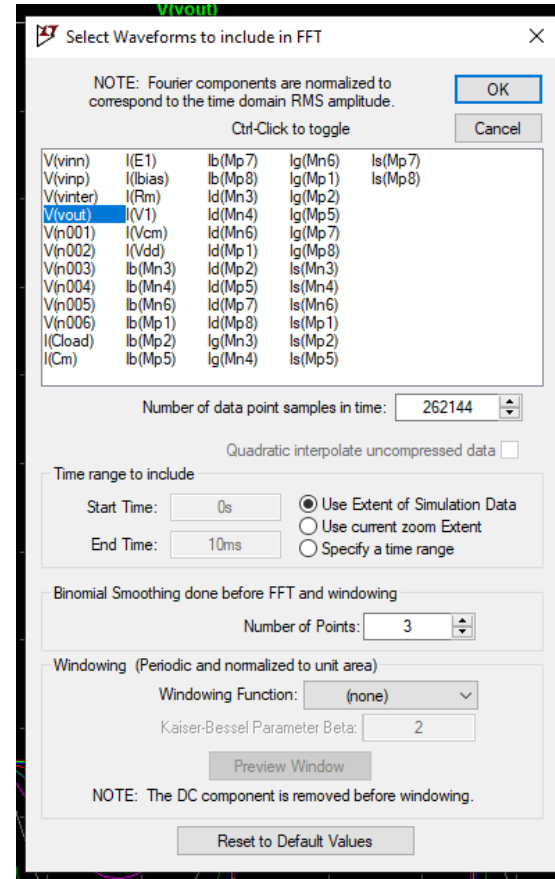


LTSPICE

Compression

NO LONGER REQUIRED

- Use the file “final_project_pss”
 - Select “V(vout)”, leave the settings as it is and press “OK” (it may take some seconds to plot)

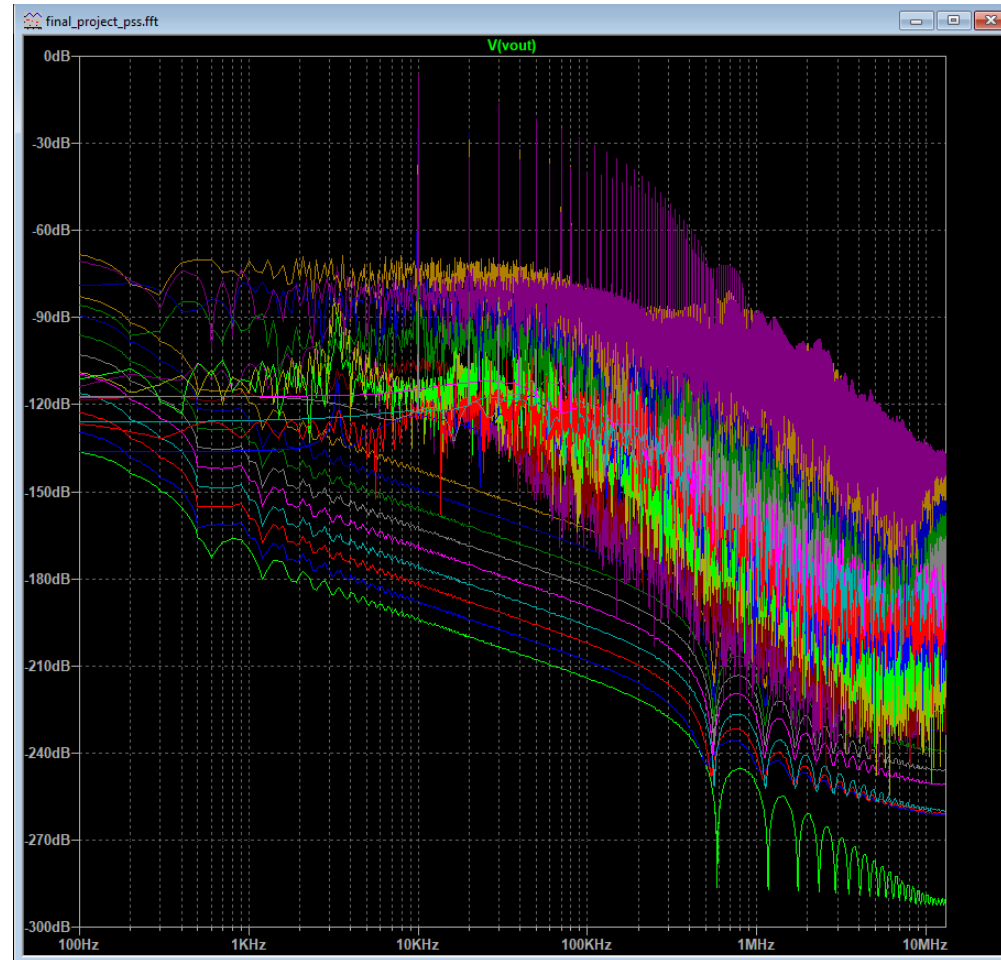


LTSPICE

Compression

NO LONGER REQUIRED

- Use the file “final_project_pss”
 - The Fourier Transform of the output signal is done for different amplitude signals

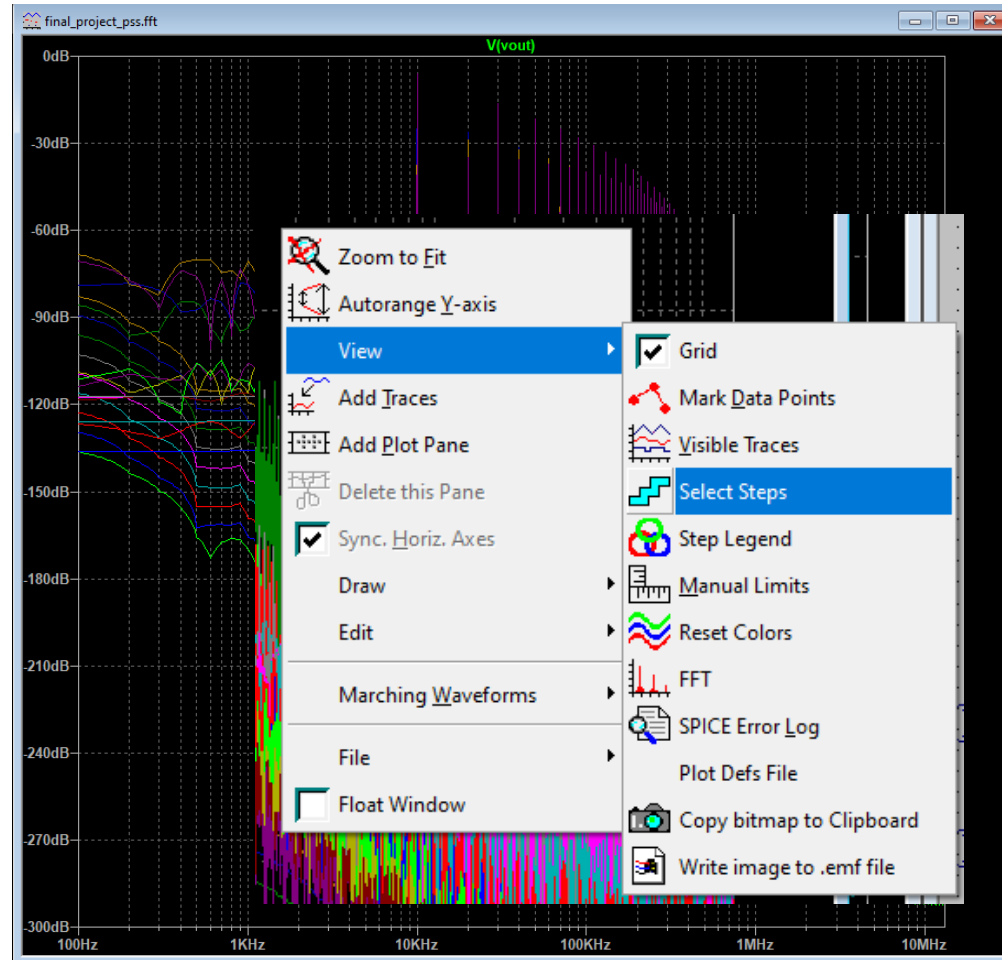


LTSPICE

Compression

NO LONGER REQUIRED

- Use the file “final_project_pss”
 - You can “Right-click” on the graph and filter by amplitude using “Select Steps”

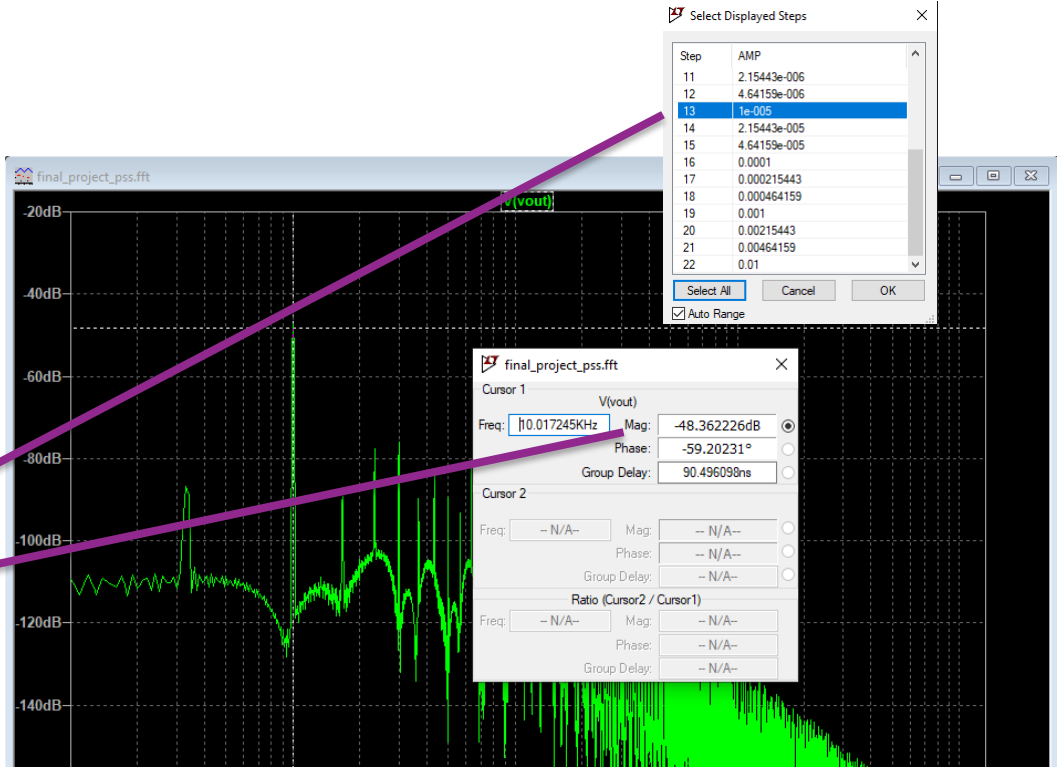


LTSPICE

Compression

NO LONGER REQUIRED

- Use the file “final_project_pss”
 - For each value of Input amplitude, find the output amplitude at 10kHz (input frequency)
 - In the figure case
 - $V_{in} = 10\mu V = -103\text{dBvrms}$
 - $V_{out} = -48.36\text{dBvrms}$
 - $\text{Gain} = -48.36 - (-100) = 51.64\text{dB}$
 - Remember: to transform from V to dBV you must use:
 - $20 \times \log_{10}(\text{value in } V)$
- Using the spreadsheet “Final_project_pss.xlsx” you just need to fill in V_{out} in dBvrms

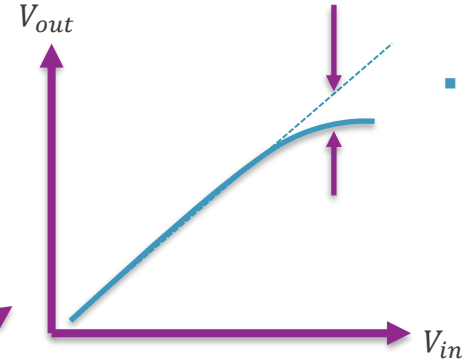


LTSPICE

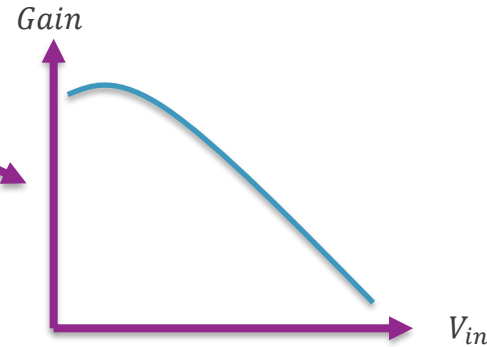
Compression

NO LONGER REQUIRED

- Plot the compression graphs
 - V_{out} (V) x V_{in} (V) in a log-log scale
 - Gain (dB) x V_{in} (V) in a semi-log scale



- Retrieve 1 dB compression point:
 - V_{in} value at which the V_{out} falls by 1 dB from the trendline

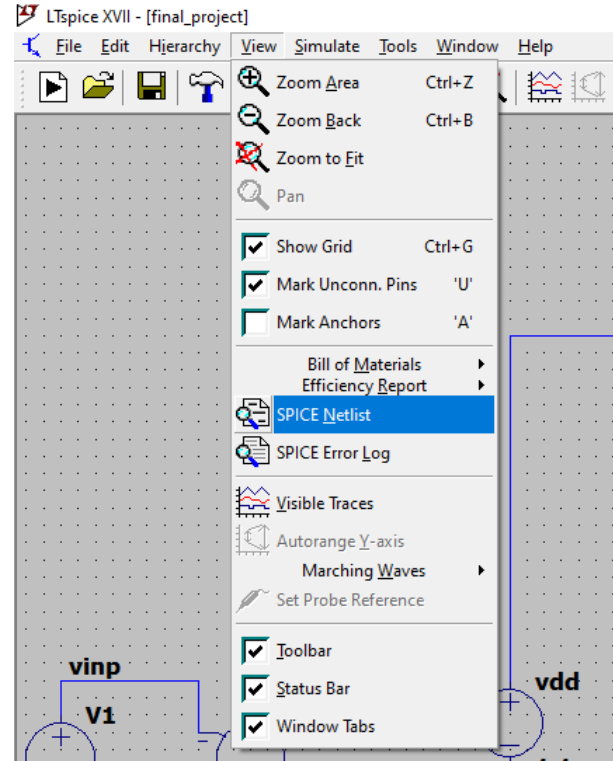


LTSPICE

Exporting the Netlist

NO LONGER REQUIRED

- On any LTSpice file after you fill in the transistor values you found, you can export the netlist following the command in the picture

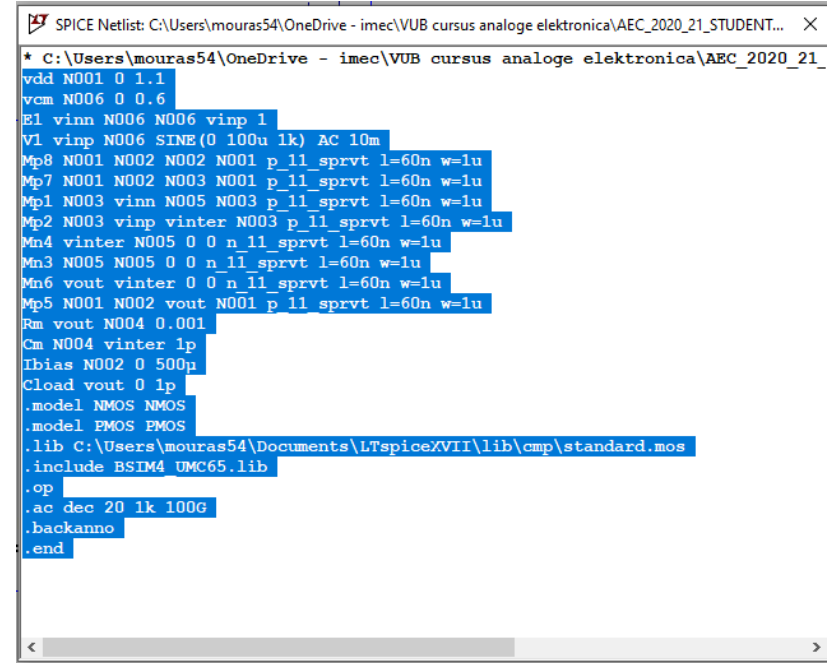


LTSPICE

Exporting the Netlist

NO LONGER REQUIRED

- Copy and paste the pop-up window in the specific part of your report



```
SPICE Netlist: C:\Users\mouras54\OneDrive - imec\VUB cursus analoge elektronica\AEC_2020_21_STUDENT... X
* C:\Users\mouras54\OneDrive - imec\VUB cursus analoge elektronica\AEC_2020_21
vdd N001 0 1.1
vcm N006 0 0.6
E1 vinn N006 N006 vinn 1
V1 vinn N006 SINE(0 100u 1k) AC 10m
Mp8 N001 N002 N002 N001 p_11_sprvt l=60n w=1u
Mp7 N001 N002 N003 N001 p_11_sprvt l=60n w=1u
Mp1 N003 vinn N005 N003 p_11_sprvt l=60n w=1u
Mp2 N003 vinn vint N003 p_11_sprvt l=60n w=1u
Mn4 vint N005 0 0 n_11_sprvt l=60n w=1u
Mn3 N005 N005 0 0 n_11_sprvt l=60n w=1u
Mn6 vout vint 0 0 n_11_sprvt l=60n w=1u
Mp5 N001 N002 vout N001 p_11_sprvt l=60n w=1u
Rm vout N004 0.001
Cm N004 vint 1p
Ibias N002 0 500p
Cload vout 0 1p
.model NMOS NMOS
.model PMOS PMOS
.lib C:\Users\mouras54\Documents\LTspice\XVII\lib\cmp\standard.mos
.include BSIM4_UMC65.lib
.op
.ac dec 20 1k 100G
.backanno
.end
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mec

embracing a better life